



DIPARTIMENTO DI ELETTRONICA,
INFORMAZIONE E BIOINGEGNERIA

Politecnico di Milano

Machine Learning (Code: 097683)

July 14, 2022

Surname:

Name:

Student ID:

Row:

Column:

Time: 2 hours

Maximum Marks: 33

- The following exam is composed of **8 exercises** (one per page). The first page needs to be filled with your **name, surname and student ID**. The following pages should be used **only in the large squares** present on each page. Any solution provided either outside these spaces or **without a motivation** will not be considered for the final mark.
- During this exam you are **not allowed to use electronic devices**, such as laptops, smartphones, tablets and/or similar. As well, you are not allowed to bring with you any kind of note, book, written scheme, and/or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above-mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above-mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time without any penalty. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.
- **Three of the points will be given on the basis on how quick you are in solving the exam.** If you finish earlier than 30 min before the end of the exam you will get 3 points, if you finish earlier than 20 min you will get 2 points and if you finish earlier than 10 min you will get 1 point (the points cannot be accumulated).
- The box on Page 10 can only be used to complete the Exercises 7, and/or 8.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Time	Tot.
/ 7	/ 7	/ 2	/ 2	/ 2	/ 2	/ 4	/ 4	/ 3	/ 33

Exercise 1 (7 marks)

Provide an overview of the **feature selection** methods that you know and explain their pros and cons.

Exercise 2 (7 marks)

Describe the **perceptron** model and how it is trained.

Exercise 3 (2 marks)

Consider the following snippet of code:

```
1 for t in range(1, T+1):
2     pulled = np.argmax(criterion == criterion.max()).reshape(-1)
3     reward = rew(pulled)
4
5     n_pulls[pulled] = n_pulls[pulled] + 1
6     exp_payoffs[pulled] = ((exp_payoffs[pulled] *
7         (n_pulls[pulled] - 1.0) + reward) / n_pulls[pulled])
8     for k in range(0, n_options)
9         criterion[k] = exp_payoffs[k] + np.sqrt(2 * t / n_pulls[k])
```

1. Describe the procedure (what algorithm it is implementing) and the purpose (which kind of problem it is solving) of the above snippet of code.
2. Is it correct? In the case the algorithm is not correct, propose a modification to fix the problem. In the case the algorithm is correct, state the theoretical guarantees of such an algorithm.
3. Are there other available methods to solve this problem? Are they requiring some specific assumptions to be applied to the same setting the algorithm in the snippet is used?

1. The code is implementing an Upper Confidence Bound based algorithm to solve the stochastic MAB problem. It uses the `criterion` variable (i.e., the upper confidence bound) to select the arm (Line 2), and, after observing the reward, it updates the number of pulls of the specific arm (Line 5), the expected payoff (Line 6) and, finally, it computes the bound (Line 9). This process is repeated over a time horizon of T rounds.
2. The algorithm is not correct since the numerator of the term under the square root should be $2 * \text{np.log}(\tau)$. Instead, this version of the bound would increase that term too much over the execution of the algorithm.
3. A viable option for the stochastic MAB problem is the Thompson Sampling algorithm, which adopts a Bayesian approach to solve the MAB scenario. The version of the algorithm we have been shown during the lectures requires that the reward are extracted from a Bernoulli distribution, otherwise the prior-posterior conjugate trick would not apply. Moreover, the other strong assumption is that the environment is stationary over time, otherwise the stochastic MAB techniques would not apply. Conversely, in that case one might use the EXP3 algorithm instead.

Exercise 4 (2 marks)

Indicates whether the following statements about Linear Regression are true or false. Motivate your answers.

1. Ordinary Least Squares is more prone to overfitting than Lasso.
2. Ridge regression performs an implicit feature selection procedure, leading to sparse solutions.
3. Lasso admits a closed-form solution of the optimal weights.
4. Both Ridge regression and Lasso introduce a bias (from a bias-variance trade-off perspective) compared to Ordinary Least Squares.

1. TRUE. Ordinary Least Squares does not employ any regularization and, therefore, is more prone to overfitting than Lasso regression that employs a regularization on the squared L_1 -norm of the weights $\|w\|_1$.
2. FALSE. Although Ridge regression employs a regularization, such a regularization is not intended to perform a feature selection and the resulting solution is not, in general, sparse. Indeed, such a property is enforced by Lasso.
3. FALSE. Lasso employs a regularization based on the L_1 -norm of the weights $\|w\|_1$. This makes the resulting objective function non differentiable and prevents from obtaining a closed-form solution. Nevertheless, the objective function remains convex and can be easily optimized numerically.
4. TRUE. They both make use of regularization techniques compared to Ordinary Least Squares. As such, from a bias-variance trade-off perspective, they increase the bias with a possible reduction of the variance.

Exercise 5 (2 marks)

Indicate whether the following statements about Monte Carlo and Temporal Difference are true or false. Motivate your answers.

1. If I am trying to evaluate a policy on a small number of interactions, it is generally better to use TD rather than MC methods.
2. MC evaluation would be a reasonable choice for providing an online estimation of a policy on a stream of data.
3. The MC every-visit evaluation suffers a smaller bias in comparison to MC first-visit evaluation.
4. The TD evaluation method is consistent, which means that it will surely converge to the optimal value function if sufficient data is available.

1. TRUE. TD difference evaluation generally suffers a smaller variance in comparison to MC methods. If I can only access a few interactions, the variance would likely be the main issue for the value estimation.
2. FALSE. MC evaluation requires full episodes to provide an estimate of the value function, which is not a good fit for an online setting. Instead, one could use TD that employs bootstrapping to provide an online estimate of the value function.
3. FALSE. MC first-visit is unbiased but suffers from a large variance, whereas MC every-visit is slightly biased but it usually reduce the variance of the estimation.
4. FALSE. TD is a consistent policy evaluation method. This means that it will converge to the exact evaluation of the policy taking the data, which can be different from the optimal policy.

Exercise 6 (2 marks)

This summer, the problem of water shortages is becoming more and more crucial in Italy as well. You are given a physical model of the water distribution network, which is able to simulate the evolution of the water reservoirs over time, providing the consumption of civil and industrial entities and the amount of water generated by rivers and other water sources.

1. Do you think the same prediction can be provided by means of ML models? Provide a modelization of the problem, the possible data you would need, and the methods you would use to solve the problem.
2. Provide a reason for preferring the developed ML model over the physical one.
3. Would you use a different ML approach based on the amount of data available?

1. This scenario can be modeled as a regression problem, where the water sources and consumer are given as input and the output is the water available in the different reservoirs. In this case, a solution might be to use Linear Regression or Gaussian Processes.
2. A motivation for which one might apply a ML model instead of a physical one is that the physical model is too simple and does not capture the complexity of the phenomenon. Another reason is that we have some uncertainty on the water distribution system that the physical model cannot include, e.g., leaks over the network.
3. On one hand, the more data is available the more complex model (e.g., computing additional features with basis functions) could be employed without fears of overfitting data. On the other hand, a large dataset could make some approaches computationally unfeasible, as Gaussian Processes, requiring to use methods that allow for incremental training, as Linear Regression.

Exercise 7 (4 marks)

Consider a linear, hard-margin, two-class SVM classifier with parameters

$$w = [2, -1], \quad b = 1.$$

Answer the following questions providing adequate motivations.

1. How is the point $p_1 = [2, 1]$ classified according to the trained SVM?
2. Is the point $p_2 = [0, 2]$ a support vector of the given SVM?
3. Provide a point that falls on the boundary of the given SVM.
4. Let us suppose we collect new data that make the classification task not linearly separable. How would you change the SVM model to deal with the task?

1. We have

$$w^\top \cdot p_1 + b = [2, -1] \cdot [2, 1]^\top + 1 = 4 - 1 + 1 = 4 \geq 1$$

thus the point is classified as positive.

2. We have

$$w^\top \cdot p_2 + b = [2, -1] \cdot [0, 2]^\top + 1 = 0 - 2 + 1 = -1$$

thus the point falls on the negative margin and it is a support vector of the SVM.

3. We look for a point x such that $w^\top \cdot x + b = 0$. We can take $x_2 = 0$ and then compute

$$[2, -1] \cdot [x_1, 0]^\top + 1 = 0 \quad \rightarrow \quad x_1 = -1/2.$$

4. We could either consider a soft-margin SVM or a different kernel function. In any case we should retrain the model to update the parameters accordingly.

Exercise 8 (4 marks)

Consider a binary classifier trained on a dataset made of $N = 100$ samples.

1. Suppose that the Precision = 0.25 and the F1 = 0.4, compute the Recall.
2. Knowing, in addition, that the Accuracy = 0.85, compute the full confusion matrix.
3. In which circumstances the Accuracy is not a reliable index to assess the quality of the trained model?

1. First of all, we compute the Recall from the F1 definition:

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \implies \text{Recall} = \frac{F1 \cdot \text{Precision}}{2 \cdot \text{Precision} - F1} = \frac{0.4 \cdot 0.25}{2 \cdot 0.25 - 0.4} = \frac{0.1}{0.1} = 1.$$

2. To get the confusion matrix, we simply apply the definition of Accuracy, Precision, and Recall, keeping in mind that the total number of samples is $N = 100$:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{TP + TN}{100} = 0.85 \implies TP + TN = 85$$

$$\text{Recall} = \frac{TP}{TP + FN} = 1 \implies FN = 0$$

$$\text{Precision} = \frac{TP}{TP + FP} = 0.25 \implies (1 - 0.25)TP = 0.25FP \implies 3TP = FP$$

Now, using $TP + TN + FP + FN = 100$, recalling that $TP + TN = 85$ and $FN = 0$, we get $FP = 15$. Using $3TP = FP$, we get $TP = 5$. Finally, using $TP + TN = 85$, we get $TN = 80$. Thus, the confusion matrix is given by:

	Actual Class: 1	Actual Class: 0
Predicted Class: 1	5	15
Predicted Class: 0	0	80

3. Accuracy is not a reliable index of the quality of the trained model mainly in two scenarios: (i) when the dataset is unbalanced; (ii) when the importance of wrongly predicting positive-class samples is different from wrongly predicting negative-class samples.

